

SMART INDIA HACKATHON 2025



LyraX

- **Problem Statement ID** - SIH25044
- **Problem Statement Title** – AI –
Powererd Crop Yield Prediction and
Optimization
- **Theme** – Agriculture, FoodTech and
Rural Development
- **PS Category** - Software
- **Team ID** – 54210
- **Team Name** - LyraX



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

✅ Proposed Solution

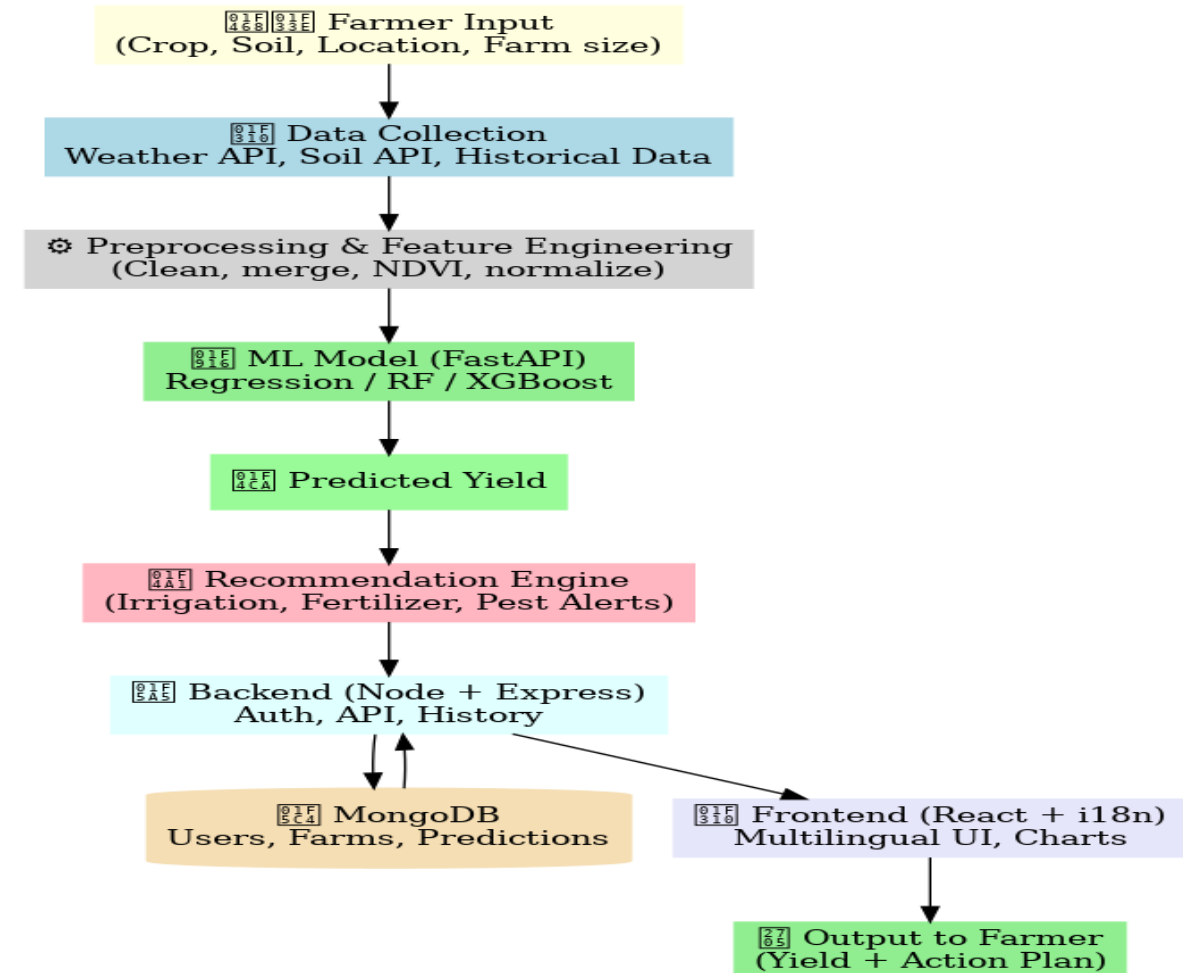
- ❑ AI-based platform that predicts crop yield using historical data, weather patterns, and soil health metrics. Recommends optimal crop selection, sowing time, fertilizers, and irrigation strategies.
- ❑ Real-time integration with weather and soil APIs for dynamic, location-specific advice.
- ❑ User-friendly, multilingual web app accessible to farmers and agri-experts across India.

🎯 How It Addresses the Problem

- ❑ Minimizes crop failure risk due to unpredictable weather.
- ❑ Boosts Reduces guesswork in farming by enabling data-driven decisions.
- ❑ Productivity by recommending best-fit practices per region.
- ❑ Empowers farmers with actionable insights in their local language.

🌟 Innovation & Uniqueness

- ❑ Combines AI, real-time data, and regional language support
- ❑ Tailored recommendations specific to crop, soil, and climate
- ❑ Accessible via simple web interface – no complex setup required
- ❑ Scalable across regions and adaptable to various crop types.



TECHNICAL APPROACH



✂ Technologies

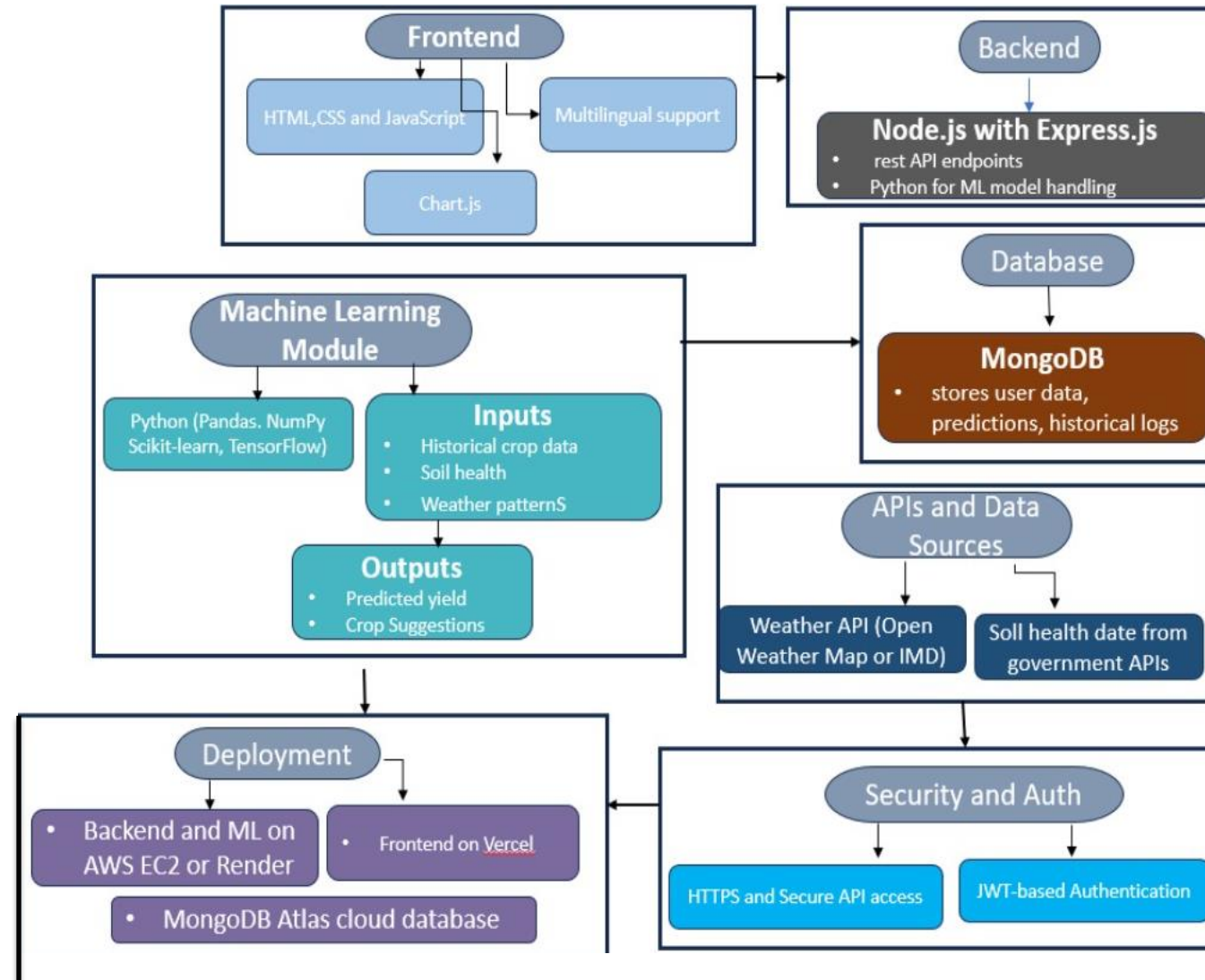
- Frontend: React.js, Tailwind CSS
- Backend: Node.js + Express, FastAPI (ML)
- ML: LightGBM (Yield), BERT (NLP), PyTorch, scikit-learn
- DB: MongoDB
- APIs: OpenWeatherMap, Soil Data API
- Version Control: GitHub

⚙ Methodology

- Data Prep: Cleaned crop, soil, weather data
- Modeling: LightGBM for yield, BERT for recos
- Integration: Live weather/soil APIs
- System: FastAPI (ML) ↔ Node.js (backend) ↔ React (UI)
- Dashboard: Predictions, charts, insights

💻 Prototype

- Input: District + Season
- Output: Yield prediction + Crop recommendations
- Features: Auto weather/soil fetch, multilingual support, farmer-friendly UI



✓ Technically Feasible:

- Built on robust MERN stack & Python ML with access to reliable weather and soil APIs.

🌍 Operationally Practical:

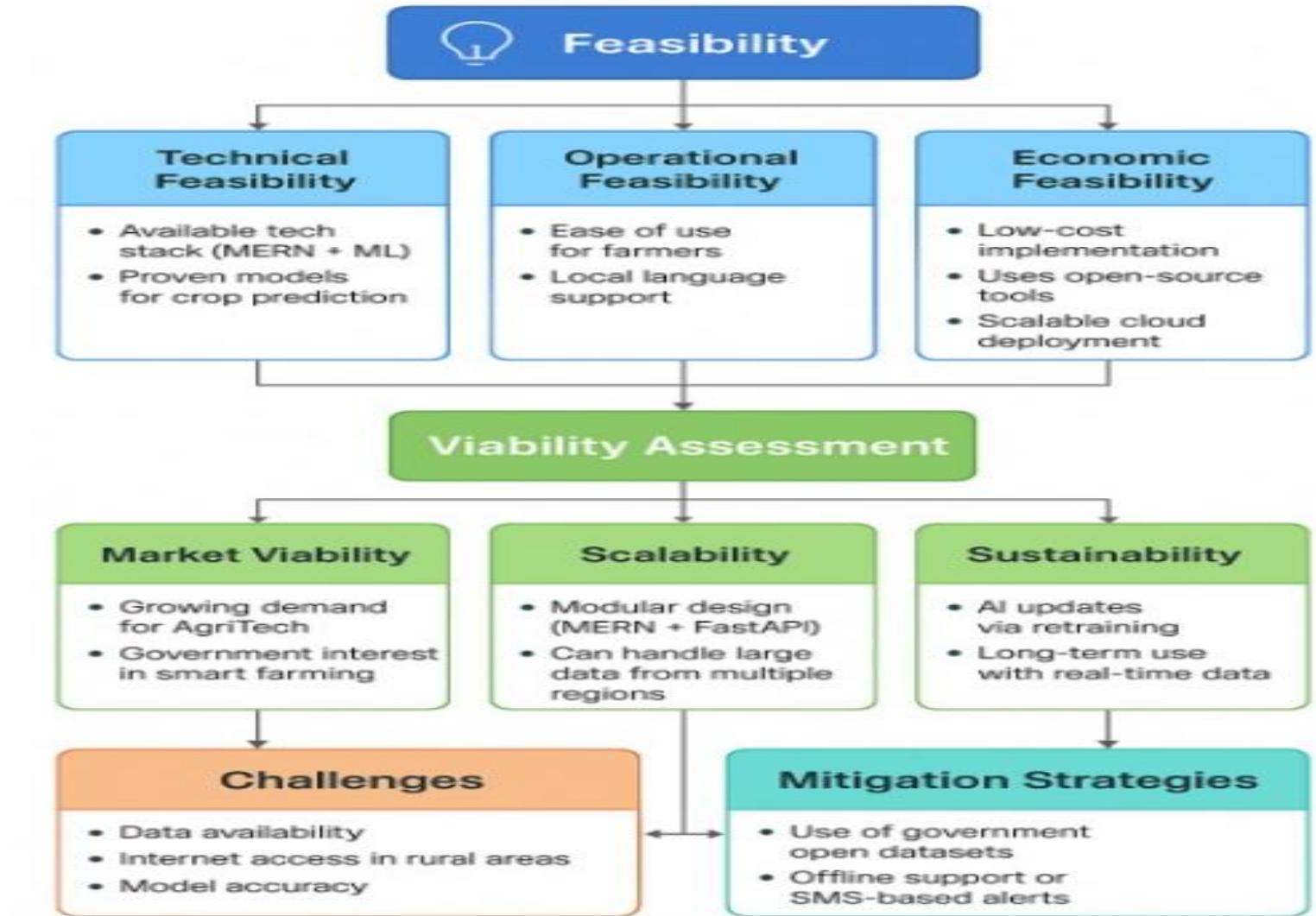
- Mobile friendly, accessible even in low- connectivity areas.

⚠️ Key Challenges:

- Incomplete agricultural data
- Low digital literacy
- Resistance to tech adoption

✂️ Our Solutions:

- Use of govt. & open datasets
- Farmer-friendly UI + regional languages to overcome low digital literacy.
- Offline mode & NGO partnerships for field outreach.
- Local language UI improves adoption in rural India.



IMPACT AND BENEFITS



Potential Impact on Farmers

- ❑ Better yield predictions → higher productivity and income.
- ❑ Data-driven decisions for irrigation, fertilization, and pest control.
- ❑ Reduced risks from weather, pests, and diseases.
- ❑ Local language support makes it accessible to all farmers.

Benefits of the Solution

Social:

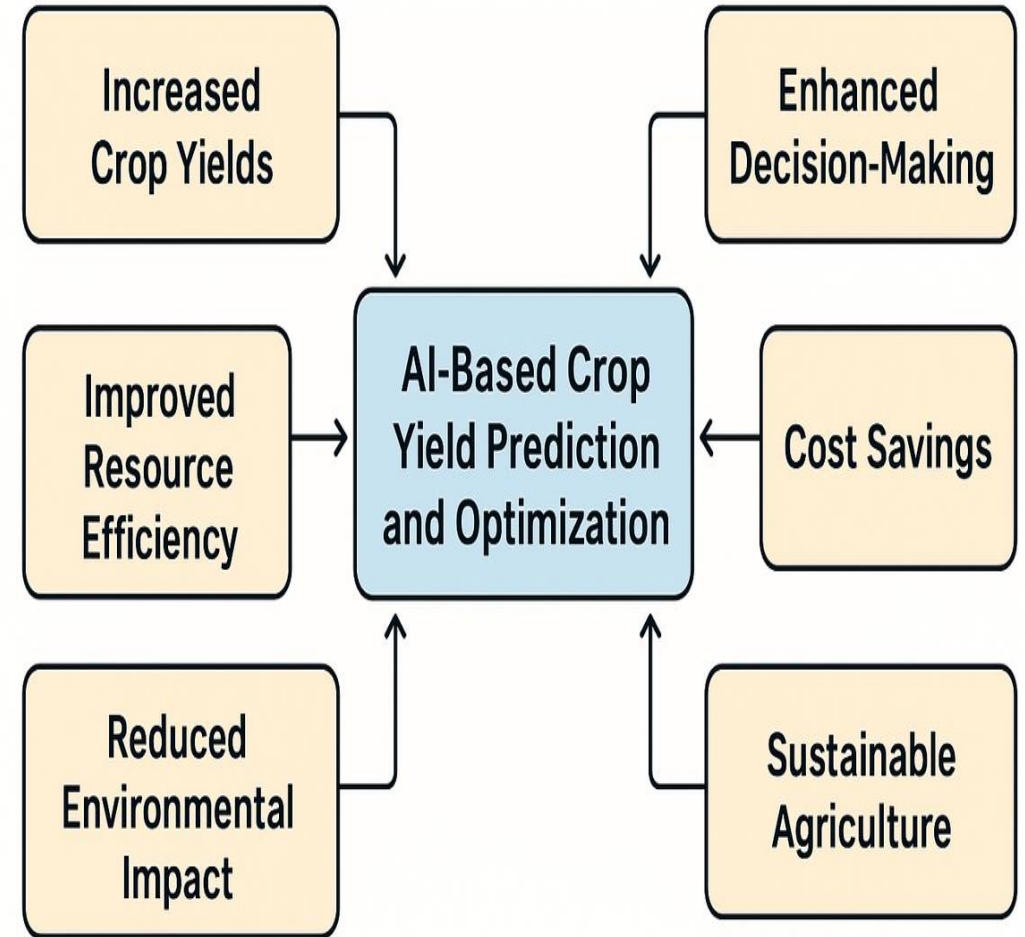
- ❑ Empowers farmers and improves food security.
- ❑ Raises awareness of best farming practices.

Economic:

- ❑ Increases income and reduces input costs.
- ❑ Improves market planning and reduces wastage.

Environmental:

- ❑ Optimizes water, fertilizer, and pesticide use.
- ❑ Promotes sustainable agriculture and protects biodiversity.



1. Research Insights:

- ☐ Studied existing crop yield prediction systems like IBM PAIRS, CropX, etc.
- ☐ Analyzed agricultural datasets from government sources (e.g., ICAR, FAO, Kaggle).
- ☐ Reviewed machine learning models used in agri-tech: Random Forest, LSTM, CNN.
- ☐ Explored problems faced by Indian farmers through news articles, surveys, and reports.

2. Sources/References:

- ☐ ICAR (Indian Council of Agricultural Research)
- ☐ FAO (Food and Agriculture Organization)
- ☐ Kaggle Datasets (Crop Yield, Rainfall, Soil Health)
- ☐ Research Papers from IEEE, Springer
- ☐ Government of India Agri Portals (e.g., agri.gov.in)
- ☐ AI/ML blogs (e.g., Towards Data Science, Medium)

3. Impact of Research:

- ☐ Helped choose optimal ML model (e.g., Random Forest over SVM).
- ☐ Identified key features affecting yield: rainfall, soil pH, temperature.
- ☐ Guided design of user-centric UI for farmers.